

1. **(10 points)** In your own words, give a detailed explanation of why “Design for Moore’s Law” is no longer one of the great ideas of computer architecture. What are some ways that designers are trying to compensate for the modern limitations of Moore’s Law? Provide an example of a piece of hardware that illustrates this.

Designing for Moore’s Law is no longer one of the great ideas of computer architecture because transistors are becoming increasingly difficult to shrink, thus making it harder to fit more transistors on a chip. Physical limits, power consumption, heat dissipation, and manufacturing costs are some of the key factors making it difficult to shrink transistors and increase clock speeds. Instead, designers have turned to multi-core systems and increased parallelism. For example, modern GPUs use thousands of smaller processing units to perform many calculations simultaneously. This illustrates the increased parallelism in modern systems that has come about thanks to somewhat ignoring Moore’s Law during design processes.

2. **(20 points)** Consider two different implementations of the same instruction set architecture. The instructions can be divided into four classes of instructions according to their CPI (class A, B, C, and D). P1 with a clock frequency of 2.7 GHz and CPIs of 2, 4, 1, and 3, and P2 with a clock frequency of 3.4 GHz and CPIS of 3, 2, 4, and 1.

Given a program with a dynamic instruction count of 1.93 E6 instructions divided into classes as follows: 20% class A, 25% class B, 35% class C, and 20% class D:

- a. What is the global CPI for each implementation?

$$CPI_{P1} = (0.2 * 2) + (0.25 * 4) + (0.35 * 1) + (0.2 * 3) = 2.35$$

$$CPI_{P2} = (0.2 * 3) + (0.25 * 2) + (0.35 * 4) + (0.2 * 1) = 2.70$$

- b. Find the clock cycles required in both cases.

$$CC_{P1} = (1.93 * 10^6)(2.35) = 4.536 * 10^6 \text{ clock cycles}$$

$$CC_{P2} = (1.93 * 10^6)(2.7) = 5.211 * 10^6 \text{ clock cycles}$$

- c. Which implementation is faster and by how much?

$$T_{P1} = \frac{4.536 * 10^6 \text{ clock cycles}}{2.7 * 10^9 \text{ cycles/sec}} = 1.680 \text{ ms}$$

$$T_{P2} = \frac{5.211 * 10^6 \text{ clock cycles}}{3.4 \text{ cycles/sec}} = 1.533 \text{ ms}$$

$$\frac{T_{P1}}{T_{P2}} = \frac{1.680}{1.533} = 1.096$$

Processor 2 is faster by about 8.76%

3. **(20 points)** Assume for arithmetic (2), load/store (9), and branch instructions (4), a processor has CPIs of 2, 9, and 4, respectively. Also assume that on a single processor, a program requires the execution of 10 E8 arithmetic instructions, 1.7 E8 load/store instructions, and 317 million branch instructions. Assume that each processor has a 3.3 GHz clock frequency.

Assume that, as the program is parallelized to run over multiple cores, the number of arithmetic instructions per processor is divided by $0.68 * p$ (where p is the number of processors) but the number of branch and load/store instructions per processor remains the same.

- a. **(10 points)** Find the total execution time for this program on 1, 2, 4, 8, and 12 processors, and show the relative speedup of the 2, 4, 8, and 12 processor result relative to the single processor result.

$$ET_1 = \frac{(10 * 10^8 * 2) + (1.7 * 10^8 * 9) + (317 * 10^6 * 4)}{3.3 * 10^9} = 1.455 \text{ s}$$

$$ET_2 = \frac{\left(\frac{10 * 10^8}{0.68 * 2} * 2\right) + (1.7 * 10^8 * 9) + (317 * 10^6 * 4)}{3.3 * 10^9} = 1.294 \text{ s}$$

$$speedup_2 = \frac{ET_1}{ET_2} = \frac{1.455 \text{ s}}{1.294 \text{ s}} = 1.124 \text{ times or } 11.03\% \text{ faster}$$

$$ET_4 = \frac{\left(\frac{10 * 10^8}{0.68 * 4} * 2\right) + (1.7 * 10^8 * 9) + (317 * 10^6 * 4)}{3.3 * 10^9} = 1.071 \text{ s}$$

$$speedup_4 = \frac{ET_1}{ET_4} = \frac{1.455 \text{ s}}{1.071 \text{ s}} = 1.359 \text{ times or } 26.42\% \text{ faster}$$

$$ET_8 = \frac{\left(\frac{10 * 10^8}{0.68 * 8} * 2\right) + (1.7 * 10^8 * 9) + (317 * 10^6 * 4)}{3.3 * 10^9} = 0.959 \text{ s}$$

$$speedup_8 = \frac{ET_1}{ET_8} = \frac{1.455 \text{ s}}{0.959 \text{ s}} = 1.517 \text{ times or } 34.09\% \text{ faster}$$

$$ET_{12} = \frac{\left(\frac{10 * 10^8}{0.68 * 12} * 2\right) + (1.7 * 10^8 * 9) + (317 * 10^6 * 4)}{3.3 * 10^9} = 0.922 \text{ s}$$

$$speedup_{12} = \frac{ET_1}{ET_{12}} = \frac{1.455 \text{ s}}{0.922 \text{ s}} = 1.578 \text{ times or } 36.63\% \text{ faster}$$

- b. **(10 points)** To what should the CPI of load/store instructions be reduced for a single processor to match the performance of four processors using the original CPI value?

$$ET_4 = 1.071 \text{ s}$$

$$CC_4 = 3.534 * 10^9$$

$$CC_1 = CC_4 = 3.534 * 10^9 = (10 * 10^8 * 2) + (1.7 * 10^8 * x) + (317 * 10^6 * 4)$$

$$3.534 * 10^9 = 3.268 * 10^9 + (1.7 * 10^8 * x)$$

$$x = \frac{3.534 * 10^9 - 3.268 * 10^9}{1.7 * 10^8} = 1.56$$

The load/store CPI should be reduced from 9 to about 1.56.

4. **(25 points)** Section 1.11 of the text cites as a pitfall the utilization of a subset of the performance equation as a performance metric. To illustrate this, consider the following two processors. P1 has a clock frequency of 3.4 GHz, average CPI of 2.4, and requires the execution of 1.8 E11 instructions. P2 has a clock frequency of 2.9 GHz, an average CPI of 0.8, and requires the execution of 0.7 E11 instructions.

- a. **(5 points)** One usual fallacy is to consider the computer with the largest clock frequency as having the largest performance. Check if this is true for P1 and P2.

$$\frac{P_1}{P_2} = \frac{IC_1 CPI_1 CF_2}{CF_1 IC_2 CPI_2} = \frac{(1.8 * 10^{11})(2.4)(2.9 * 10^9)}{(3.4 * 10^9)(0.7 * 10^{11})(0.8)} = 6.580$$

This is not true for P1 and P2. As calculated above, P2 has a better performance even though it has a smaller clock frequency.

- b. **(10 points)** Another fallacy is to consider that the processor executing the largest number of instructions will need a larger CPU time. Considering that processor P1 is executing a sequence of 2.5 E9 instructions and that the CPI of processors P1 and P2 do not change, determine the

number of instructions that P2 can execute in the same time that P1 needs to execute 2.5×10^9 instructions.

$$ET_1 = \frac{(2.5 * 10^9)(2.4)}{3.4 * 10^9} = 1.765 \text{ s}$$

$$\text{effective } ET_2 = 1.765 \text{ s} = \frac{IC_2 CPI_2}{CF_2} = \frac{IC_2(0.8)}{2.9 * 10^9}$$

$$IC_2 = \frac{(1.765)(2.9 * 10^9)}{0.8} = 6.398 * 10^9$$

Processor 2 executes $6.398 * 10^9$ instructions in the same amount of time it takes Processor 1 to execute $2.5 * 10^9$ instructions. Therefore, the processor executing more instruction does not always require more execution time.

- c. **(10 points)** A common fallacy is to only use MIPS (millions of instructions per second) to compare the performance of two different processors, and consider that the processor with the largest MIPS has the largest performance. Check if this is true for P1 and P2.

$$MIPS_1 = \frac{CF_1 * 1000}{CPI} = \frac{3.4 * 1000}{2.4} = 1416.667 \text{ MIPS}$$

$$MIPS_2 = \frac{CF_2 * 1000}{CPI} = \frac{2.9 * 1000}{0.8} = 3625 \text{ MIPS}$$

Processor 2 has the higher MIPS rate, though it has the smaller performance. This means that the processor with the highest MIPS rate does not always have the largest performance.

5. **(5 points)** Another pitfall cited in Section 1.11 is expecting to improve the overall performance of a computer by improving only one aspect of the computer. Consider a computer running a program that requires 380 s, with 90 s spent executing FP instructions, 115 s executing L/S instructions, and 50 s spent executing branch instructions. Can the total time be reduced by 20 % by reducing only the time for L/S instructions?

$$ET = 380 \text{ s}$$

$$0.8 * ET = 304 \text{ s}$$

$$LS \text{ reduction} = \frac{76}{115} * 100 = 66.086\%$$

Since the execution time needs to be reduced by 76 seconds to get an overall reduction of 20%, it is possible to reduce the total time by reducing only the L/S instructions. This would be challenging, however, as the number of L/S instructions would need to be reduced by about 66%.

6. **(15 points)** Assume a program requires the execution of 70 E8 FP instructions, 120 E8 INT instructions, 73 E8 L/S instructions, and 12 E6 branch instructions. The CPI for each type of instruction is 12, 3, 6, and 4, respectively. Assume that the processor has a 4.2 GHz clock frequency.

- a. **(10 points)** By how much must we improve the CPI of INT instructions if we want the program to run two times faster?

$$CC_{orig} = (70 * 10^8 * 12) + (120 * 10^8 * 3) + (73 * 10^8 * 6) + (12 * 10^6 * 4)$$

$$= 163.848 * 10^9 \text{ clock cycles}$$

$$ET_{orig} = \frac{163.848 * 10^9}{4.2 * 10^9} = 39.011 \text{ s}$$

$$CC_{target} = \frac{1}{2} CC_{orig} = 81.924 * 10^9 \text{ clock cycles}$$

$$CC_{nonINT} = (70 * 10^8 * 12) + (73 * 10^8 * 6) + (12 * 10^6 * 4) = 1.27.848 * 10^9 \text{ clock cycles}$$

Since CC_{nonINT} exceeds the number of clock cycles needs to reach a 200% speedup, it is impossible to reach the desired performance while only reducing the number of INT instructions.

- b. (5 points) By how much is the execution time improved if the CPI of INT and FP instructions is reduced by 32% and the CPI of L/S and Branch is reduced by 19%?

$$CC = (70 * 10^8 * 12 * 0.68) + (120 * 10^8 * 3 * 0.68) + (73 * 10^8 * 6 * 0.81) + (12 * 10^6 * 4 * 0.81) = 117.117 * 10^9 \text{ cycles}$$

$$ET = \frac{117.117 * 10^9}{4.2 * 10^9} = 27.885 \text{ s}$$

$$ET_{reduct} = \frac{39.011 - 27.885}{39.011} * 100 = 28.520\%$$

The execution time is improved by 28.520% or 1.399 times.

7. (10 points) When a program is adapted to run on multiple processors in a multiprocessor system, the execution time on each processor is comprised of computing time and the overhead time required for locked critical sections and/or to send data from one processor to another.

Assume a program requires $t = 250$ s of execution time on one processor. When run on p processors, each processor requires t/p seconds, as well as an additional 10 s of overhead, irrespective of the number of processors. Compute the per-processor execution time for 2, 4, 8, 16, 32, 64, and 128 processors. For each case, list the corresponding speedup relative to a single processor and the ratio between actual speedup versus ideal speedup (speedup if there was no overhead).

Number of processors	$T_p = \frac{250}{p} + 10$	$S_p = \frac{250}{T_p}$	$T_{ideal} = \frac{250}{p}$	$S_{ideal} = p$	$\frac{S_{actual}}{S_{ideal}}$
2	135 s	1.852	2	2	0.926
4	72.5 s	3.448	4	4	0.862
8	41.25 s	6.061	8	8	0.758
16	25.625 s	9.756	16	16	0.610
32	17.813 s	14.035	32	32	0.439
64	13.906 s	17.978	64	64	0.281
128	11.953 s	20.915	128	128	0.164

8. (15 points) The Pentium 4 Prescott processor from 2004 had a clock rate of 3.6 GHz, and an operating voltage of 1.25 V. Assume that, on average, it consumed 10 Watts of static power and 90 Watts of dynamic power.

The Core i5 Ivy Bridge from 2012 has a clock rate of 3.4 GHz and an operating voltage of 0.9V. Assume that, on average, it consumed 30W of static power and 40W of dynamic power.

- a. (2 points) What is the average capacitive load for each processor?

$$\text{Capacitive Load (CL)} = \frac{P_{dyn}}{V^2 f}$$

$$CL_{Prescott} = \frac{90}{1.25^2 * (3.6 * 10^9)} = 16 \text{ nF}$$

$$CL_{Ivy} = \frac{40}{0.9^2 * (3.4 * 10^9)} = 14.51 \text{ nF}$$

- b. (5 points) What is the percentage of the total dissipated power due to static power and the ratio of static to dynamic power for each technology node?

$$P_{stat,diss,Prescott} = \frac{10}{100} = 10\%$$

$$power\ ratio_{Prescott} = \frac{10}{90} = 11.111\%$$

$$P_{stat,diss,Ivy} = \frac{30}{70} = 42.86\%$$

$$power\ ratio_{Ivy} = \frac{30}{40} = 75\%$$

- c. (8 points) If we wish to reduce dissipated power by 10%, how much should the voltage be reduced to maintain the same leakage current?

$$P_{new,Prescott} = 90W$$

$$P_{static,new,Prescott} = P_{static,old} \left(\frac{V_{new}}{V_{old}} \right) = 10 \left(\frac{V_{new}}{1.25} \right)$$

$$P_{dyn,new,Prescott} = P_{dyn,old} \left(\frac{V_{new}}{V_{old}} \right)^2 = 90 \left(\frac{V_{new}}{1.25} \right)^2$$

$$P_{static,new,Prescott} + P_{dyn,new,Prescott} = 90W$$

$$\text{Let } x = \frac{V_{new}}{1.25}$$

$$10x + 90x^2 = 90$$

$$x = \frac{-1 + 18.028}{18} = 0.946$$

$$V_{new} = 1.25x = 1.183V$$

$$reduction_{Prescott} = \frac{1.25 - 1.183}{1.25} * 100 = 5.26\% \text{ reduction in voltage}$$

$$P_{new,Ivy} = 63W$$

$$P_{static,new,Prescott} = P_{static,old} \left(\frac{V_{new}}{V_{old}} \right) = 30 \left(\frac{V_{new}}{0.9} \right)$$

$$P_{dyn,new,Prescott} = P_{dyn,old} \left(\frac{V_{new}}{V_{old}} \right)^2 = 40 \left(\frac{V_{new}}{0.9} \right)^2$$

$$P_{static,new,Prescott} + P_{dyn,new,Prescott} = 63W$$

$$\text{Let } x = \frac{V_{new}}{0.9}$$

$$30x + 40x^2 = 63$$

$$x = \frac{-30 + 104.785}{80} = 0.945$$

$$V_{new} = 0.9x = 0.841V$$

$$reduction_{Prescott} = \frac{0.9 - 0.841}{0.9} * 100 = 6.56\% \text{ reduction in voltage}$$

To maintain the same leakage current, the voltage for both the Prescott and Ivy devices should be reduced by 5.26% and 6.56% respectively.

9. (15 points) Assume a 15 cm diameter wafer has a cost of 12, contains 84 dies, and has 0.020 defects/cm². Assume a 20 cm diameter wafer has a cost of 15, contains 100 dies, and has 0.031 defects/cm².

- a. (2.5 points) What is the yield of each wafer?

$$yield_{15cm} = \frac{1}{\left(1 + \left(0.02 * \frac{\pi 7.5^2}{84}\right)\right)^2} = 92.09\%$$

$$yield_{20cm} = \frac{1}{\left(1 + \left(0.031 * \frac{\pi 10^2}{100}\right)\right)^2} = 83.04\%$$

- b. (2.5 points) What is the cost per die for each wafer?

$$CPD_1 = \frac{12}{84 * 0.9209} = \$0.155$$

$$CPD_2 = \frac{15}{100 * 0.8304} = \$0.186$$

- c. (5 points) Determine the die area and yield for each wafer if number of dies per wafer is increased by 10% and the defects per area increases by 15%.

$$Die\ area = DA \approx \frac{Wafer\ area}{Dies\ per\ wafer}$$

$$DA_{15cm,new} = \frac{\pi 7.5^2}{1.1(84)} = 1.912\ cm^2$$

$$yield_{15cm,new} = \frac{1}{\left(1 + (0.023 * 1.912)\right)^2} = 91.75\%$$

$$DA_{20cm,new} = \frac{\pi 10^2}{1.1(100)} = 2.856\ cm^2$$

$$yield_{20cm,new} = \frac{1}{\left(1 + (0.03565 * 2.856)\right)^2} = 82.37\%$$

- d. (5 points) Find the before and after defects per area unit for a 200 mm² die that goes through a process that's yield has improved from 0.92 to 0.95.

$$D = \frac{\frac{1}{\sqrt{Y}} - 1}{A}$$

$$200\ mm^2 = 2\ cm^2$$

$$D_{before} = \frac{\frac{1}{\sqrt{0.92}} - 1}{2} = 0.0213\ defects/cm^2$$

$$D_{after} = \frac{\frac{1}{\sqrt{0.95}} - 1}{2} = 0.0130\ defects/cm^2$$

10. (15 points) The results of the SPEC CPU2006 bzip2 benchmark running on an AMD Barcelona processor has an instruction count of 2.389E12, an execution time of 750 s, and a reference time of 9650 s.

- a. **(2.5 points)** What is the CPI if the clock speed is 0.333 ns?

$$CPI = \frac{ET}{IC * CS} = \frac{750}{(2.389 * 10^{12})(0.333 * 10^{-9})} = 0.943$$

- b. **(2.5 points)** What is the SPECratio for this benchmark?

$$SPECratio = \frac{RT}{ET} = \frac{9650}{750} = 12.867$$

- c. **(2.5 points)** If the number of instructions of the benchmark is increased by 10% without affecting the CPI, what is the new CPU time? What is the CPU time if the CPI is also increased by 5%?

$$ET_{new\ IC} = 750(1.1) = 825\ s$$

$$ET_{new\ IC\ and\ CPI} = 750(1.1)(1.05) = 866.25\ s$$

- d. **(2.5 points)** What is the new SPECratio for the 10% change in instruction count and the 5% change in CPI?

$$SPECratio_{new\ IC} = \frac{RT}{ET} = \frac{9650}{825} = 11.697$$

$$SPECratio_{new\ IC\ and\ CPI} = \frac{RT}{ET} = \frac{9650}{866.25} = 11.140$$

- e. **(5 points)** What change would need to be made to the clock rate to achieve the SPECratio from part b with the change in instruction count and CPI from part d?

$$speedup = \frac{866.25}{750} = 1.155\ or\ 15.5\%\ faster$$

$$f_{orig} = \frac{1}{(0.333 * 10^{-9})} = 3.003\ GHz$$

$$\frac{f_{new}}{f_{orig}} = 1.115$$

$$f_{new} = 1.115(3.003 * 10^9) = 3.468\ GHz$$

To achieve the same SPECratio as seen in part b, the clock rate would need to increase by 15.5% (or from 3.003 GHz to 3.468 GHz).